Supporting information

Population-size fidelity in reference-vector multi-objective evolutionary algorithms

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Table S1. Static settings

problem	n_{ex}=5, γ=0	n_{ex}=5, γ=0.7	n_{ex}=5, γ=0.3	n_{ex}=10, γ=0.7	n_{ex}=20, γ=0	best
DTLZ1	0.0515	0.0422	0.0481	0.0420	0.0737	n_{ex}=10, γ=0.7
DTLZ2	0.0542	0.0531	0.0595	0.0540	0.0650	n_{ex}=5, γ=0.7
DTLZ7	0.0569	0.0518	0.0569	0.0499	0.0570	n_{ex}=10, γ=0.7
LSMOP1	0.0840	0.0898	0.0864	0.0958	0.1151	n_{ex}=5, γ=0
LSMOP2	0.0829	0.0911	0.0867	0.0907	0.0793	n_{ex}=20, γ=0
LSMOP3	0.8636	0.8636	0.8635	0.8640	0.8644	n_{ex}=5, γ=0.3
LSMOP4	0.1436	0.1417	0.1463	0.1435	0.1487	n_{ex}=5, γ=0.7
LSMOP5	0.3509	0.3567	0.3549	0.3566	0.3586	n_{ex}=5, γ=0
LSMOP6	0.6360	0.6471	0.6367	0.6490	0.6529	n_{ex}=5, γ=0
LSMOP7	0.8116	0.8593	0.8274	0.8589	0.7953	n_{ex}=20, γ=0
LSMOP8	0.0737	0.0770	0.0743	0.0807	0.0818	n_{ex}=5, γ=0
LSMOP9	0.7343	0.7346	0.7347	0.7352	0.7349	n_{ex}=5, γ=0
LSMOP1 (5,500)	0.3759	0.3518	0.3708	0.3515	0.4031	n_{ex}=10, γ=0.7
LSMOP6 (5,500)	0.8857	0.9835	0.8858	1.0116	0.8974	n_{ex}=5, γ=0
LSMOP1 (3,1000)	0.0799	0.0926	0.0893	0.1137	0.1144	n_{ex}=5, γ=0

Median IGD of the host under five static settings of the exchange count n_{ex} and the hybridisation weight γ, on the fifteen problems, with the best setting per problem (the "best" column names it). This sweep supplies the reference against which the online scheduling layer of Section 5.5 is judged.

Table S2. Sensitivity to the exchange count

problem	K=3	K=5	K=10	best
DTLZ1	0.0418	0.0420	1116.0403	K=3
DTLZ2	0.0537	0.0540	0.0625	K=3
DTLZ7	0.0495	0.0499	0.0613	K=3
LSMOP1	0.0942	0.0958	0.1032	K=3
LSMOP2	0.0903	0.0907	0.0975	K=3
LSMOP3	0.8645	0.8640	0.8608	K=10
LSMOP4	0.1479	0.1435	0.1405	K=10
LSMOP5	0.4216	0.3566	0.3488	K=10
LSMOP6	0.6941	0.6490	0.7841	K=5
LSMOP7	0.8902	0.8589	0.8320	K=10
LSMOP8	0.0824	0.0807	0.0882	K=5
LSMOP9	0.7357	0.7352	0.7345	K=10
LSMOP1 (5,500)	0.3788	0.3515	0.3486	K=10
LSMOP6 (5,500)	1.0356	1.0116	1.5016	K=5
LSMOP1 (3,1000)	0.1136	0.1137	0.1022	K=10

Median IGD when the exchange count K is swept over $\{3, 5, 10\}$ at the fixed setting $n_{ex} = 10$, $\gamma = 0.7$, with the best value per problem. The $K = 5$ column reproduces the $(10, 0.7)$ column of Table S1, which fixes the reference arm of the sweep.

Table S3. Repair 1 and repair 2

problem	published	repair 1	repair 2	Δ_1	Δ_2
DTLZ1	0.0657	0.0584	0.0737	+11.1%	−12.1%
DTLZ2	0.0609	0.0603	0.0658	+1.0%	−7.9%
DTLZ7	0.0590	0.0543	0.0587	+7.9%	+0.4%
LSMOP1	0.1032	0.0945	0.1030	+8.4%	+0.2%
LSMOP2	0.0812	0.0823	0.0866	−1.5%	−6.7%
LSMOP3	0.8645	0.8645	0.8646	+0.0%	−0.0%
LSMOP4	0.1448	0.1424	0.1457	+1.7%	−0.6%
LSMOP5	0.3559	0.3531	0.3580	+0.8%	−0.6%
LSMOP6	0.6429	0.6363	0.6427	+1.0%	+0.0%
LSMOP7	0.8052	0.8383	0.8852	−4.1%	−9.9%
LSMOP8	0.0769	0.0748	0.0783	+2.7%	−1.9%
LSMOP9	0.7359	0.7351	0.7380	+0.1%	−0.3%
LSMOP1 (5,500)	0.3971	0.3736	0.3790	+5.9%	+4.6%
LSMOP6 (5,500)	0.9400	0.8693	0.8822	+7.5%	+6.1%
LSMOP1 (3,1000)	0.0992	0.0904	0.1039	+8.9%	−4.8%

Median IGD of the published arm and of the two admissible repairs, with the median relative change of each repair against the published arm (Δ_1, Δ_2; positive means the repair attains a smaller indicator).

Table S4. The bounded-angle constructor (Case C)

N	M	lattice L	final size	$\gamma_{\min}$
100	3	91	100	5.19°
100	5	85	100	18.19°
100	8	72	100	16.38°

Requested size N, number of objectives M, the size L of the underlying lattice, the final size and the minimum inter-vector angle $\gamma_{\min}$ of the bounded-angle constructor. The floor $\theta_{\min} = 1^\circ$ was never binding on the settings studied.

Table S5. The scheduling layer

problem	published	requested N	layer	vs published	vs requested
DTLZ1	0.0657	0.0584	0.0421	+36.0%	+28.0%
DTLZ2	0.0609	0.0603	0.0528	+13.3%	+12.4%
DTLZ7	0.0590	0.0543	0.0506	+14.2%	+6.8%
LSMOP1	0.1032	0.0945	0.0872	+15.5%	+7.7%
LSMOP2	0.0812	0.0823	0.0862	−6.3%	−4.7%
LSMOP3	0.8645	0.8645	0.8642	+0.0%	+0.0%
LSMOP4	0.1448	0.1424	0.1424	+1.7%	−0.0%
LSMOP5	0.3559	0.3531	0.3528	+0.9%	+0.1%
LSMOP6	0.6429	0.6363	0.6572	−2.2%	−3.3%
LSMOP7	0.8052	0.8383	0.8456	−5.0%	−0.9%
LSMOP8	0.0769	0.0748	0.0793	−3.1%	−6.0%
LSMOP9	0.7359	0.7351	0.7352	+0.1%	−0.0%
LSMOP1 (5,500)	0.3971	0.3736	0.3448	+13.2%	+7.7%
LSMOP6 (5,500)	0.9400	0.8693	1.0206	−8.6%	−17.4%
LSMOP1 (3,1000)	0.0992	0.0904	0.0893	+9.9%	+1.2%

Median IGD of the scheduling layer against the published host and against the host at the requested population size, with the median relative change against each reference.

Table S6. Declared-versus-published p-values

problem	p (declared vs published)	verdict
DTLZ1	0.6420	no separation
DTLZ2	0.9455	no separation
DTLZ7	0.7570	no separation
LSMOP1	0.1909	no separation
LSMOP2	0.1519	no separation
LSMOP3	1.0000	no separation
LSMOP4	0.3818	no separation
LSMOP5	0.6417	no separation
LSMOP6	0.1394	no separation
LSMOP7	0.5337	no separation
LSMOP8	0.7775	no separation
LSMOP9	0.8589	no separation
LSMOP1 (5,500)	0.7012	no separation
LSMOP6 (5,500)	0.1231	no separation
LSMOP1 (3,1000)	0.8313	no separation

Two-sided paired Wilcoxon signed-rank p-values of the declared arm against the published arm of Section 5.2 (the two same-size arms). None separates, which is why the difference between them is reported as an unresolved residual.

Table S7. Catalogue validation on DTLZ1–7

problem	MOEA/D		RVEA			LMOCSO		
	pub	repair 2	pub	repair 2	bounded	pub	repair 2	bounded
DTLZ1	0.0401	0.0412	0.0400	0.0413	0.0390	0.0443	0.0444	0.0442
DTLZ2	0.0522	0.0532	0.0522	0.0532	0.0505	0.0523	0.0532	0.0505
DTLZ3	0.0525	0.0535	0.0523	0.0534	0.0507	5.8193	6.3026	5.9530
DTLZ4	0.0522	0.0532	0.0522	0.0532	0.0505	0.5409	0.5405	0.3962
DTLZ5	0.0403	0.0158	0.0740	0.2322	0.0606	0.0393	0.9196	0.0404
DTLZ6	0.0403	0.0162	0.1390	0.3400	0.0968	0.0401	0.9583	0.0403
DTLZ7	0.1023	0.0906	0.0649	0.0581	0.0640	0.3209	0.3238	0.3206

Median IGD at three objectives (requested $N = 100$, budget 10^5 evaluations) of the published, repair-2 and bounded-angle variants for MOEA/D, RVEA and LMOCSO. Repair 2 restores MOEA/D to exactly 100 individuals; the bounded-angle constructor repairs the Case-C pairs on the degenerate fronts DTLZ5–6.

Table S8. The indicator-cardinality IGD control

problem	raw (%)	indicator-card., random (%)	HV raw (%)
DTLZ1	+11.1	+7.0	+0.54
DTLZ2	+1.0	−5.6	+0.09
DTLZ7	+7.9	+4.5	+1.57
LSMOP1	+8.4	+6.2	+0.46
LSMOP2	−1.5	−4.4	−0.04
LSMOP3	0.0	0.0	0.00
LSMOP4	+1.7	+0.5	+0.34
LSMOP5	+0.8	+0.8	+0.00
LSMOP6	+1.0	+1.0	−0.00
LSMOP7	−4.1	−4.1	−0.45
LSMOP8	+2.7	−1.7	+0.62
LSMOP9	+0.1	+0.1	+0.12
LSMOP1 $(5, 500)$	+5.9	+4.5	+0.01
LSMOP6 $(5, 500)$	+7.5	+7.0	+0.00
LSMOP1 $(3, 1000)$	+8.9	+6.4	+0.55
median	+1.7	+0.8	+0.09
better than published	12/15	10/15	13/15

Median relative change of the repaired host against the published host ("raw"), and the same change after each repaired front is downsampled to the published arm's realised size (91 at three objectives, 85 at five) so that the two arms are compared at equal cardinality. The downsampled indicator is a two-level median: first over 200 subsets of that size drawn uniformly without replacement (NumPy `default_rng(0)`) within each run, then over the 30 (20) runs, recomputed against the same reference front (PlatEMO `GetOptimum(400)`, reconstructed in Python). A coverage-maximising (farthest-point) subset is deliberately not used: it optimises the spread of the selected points rather than the distance-to-front that IGD averages and can select outliers that raise IGD. The last column gives the same raw comparison under the exact hypervolume (normalised objectives, reference point at 1.1). The bottom rows give the median change over the fifteen problems and the number on which the repaired host is better; the two indicators agree in sign on 13 of the 15 problems.

Table S9. Trace of the three quantities

algorithm	N_{req}	L	N_{real}	N_{rep}	status
NSGA-III	100	91	91	91	diverges
MOEA/D	100	91	91	91	diverges
RVEA	100	91	83	91	diverges
LMOCSO	100	91	66	91	diverges
FDSEA	100	91	91	91	diverges
HEA	100	91	91	91	diverges
C-TAEA	100	91	91	91	diverges
MOEA/D-DRA	100	91	91	91	diverges
MOEA/D-DE	100	91	91	91	diverges
RVEAa	100	91	31	91	diverges
DGEA	100	91	80	91	diverges
PPS	100	91	91	91	diverges
MOEA/D-STM	100	91	91	91	diverges
NSGA-II	100	–	100	100	faithful
SPEA2	100	–	100	100	faithful
IBEA	100	–	100	100	faithful

Requested, lattice, realised and reported sizes for the traced reference-vector algorithms, all configured with $N_{\mathrm{req}} = 100$ at $M = 3$; the last column gives the status of each algorithm.

Table S10. Realised count on the grid

$M \setminus N$	91	100	105	120	150	200	275	300	500
2	**91**	**100**	**105**	**120**	**150**	**200**	**275**	**300**	**500**
3	**91**	91	**105**	**120**	136	190	253	**300**	496
4	84	84	84	**120**	120	165	220	286	455
5	85	85	**105**	105	126	126	210	210	495
6	77	77	77	112	147	182	273	273	462
7	**91**	91	91	112	112	168	238	294	490
8	72	72	72	**120**	128	156	240	240	450
9	90	90	90	90	90	174	210	210	495
10	65	65	65	110	110	110	**275**	275	440

Realised count L for each requested size and objective count on the grid used in Section 3.1.